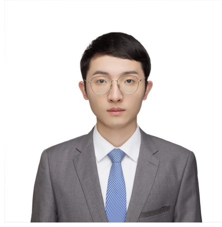
Jingwei Cai



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Education

Tsinghua University, Institute for Interdisciplinary Information Sciences, PhD in Computer Systems Architecture, *Advisor: Kaisheng Ma* 2021.9 - 2026.6

- **Research Interests**: DNN accelerator architecture and compiler design based on 2.5D/3D/wafer-scale Chiplets; Wafer-scale silicon photonic interconnected DNN computing systems; Recommendation system acceleration
- **Funding Awards**: **China Association for Science and Technology Young Talent Cultivation Program - Doctoral Special Project (only 1 Ph.D. student selected from IIIS)**; Electronics Society Talent Cultivation Award; Student Member Funding from Electronics Society (20 selected nationwide); Total funding: 45K RMB
- **Publications**: 7 papers published, 5 of which as first author in CCF-A tier computer architecture top conferences; Recipient of **HPCA 2024 Distinguished Artifact Award (1/410, first time for Chinese researchers)**; ISCA 2023, HPCA 2024 Travel Grants
- **Scholarships**: **Wang Dazho Scholarship (Tsinghua Super Scholarship candidate, 20 graduate students university-wide)**; **Doctoral National Scholarship** (only 2 CS Ph.D. students per year from IIIS); **ByteDance Scholarship (20 selected from China and Singapore annually)**; Yangtze River Delta International Talent Scholarship; Beijing Outstanding Graduates; Institute Scholarships
- **Academic Service**: Reviewer for CCF-B IEEE Computer Architecture Letters (CAL), CCF-B JCST, IEEE Transactions on Sustainable Energy (IF=10); Sub-reviewer for CCF-A HPCA 2025, ISCA 2025, and TCAD journal
- **Invited Talks**: DataFunSummit 2024: AI Foundational Software Architecture Summit; Shanghai Cooperation Organization Summit Doctoral Innovation Forum: Future AI Accelerators
- Class Monitor of IIIS Graduate Class 2; Minister of Activities in IIIS Graduate Association (awarded 2021-2022 Outstanding Graduate Star); Team Leader for December 9th Thematic Education Activity (broke historical records for IIIS, won Comprehensive Second Prize; selected as December 9th Star and Excellent Team Leader)

University of Electronic Science and Technology of China, School of Electronic Engineering, BS in Integrated Circuit Design and Integration Systems, *Advisor: Letian Huang* 2017.9 - 2021.6

- **National Scholarship** * 3 (selected as **2021 People's Daily National Scholarship Representative, one of 100 nationwide**); **UESTC Distinguished Student** (10 selected per year from graduates, equivalent to **Tsinghua Distinguished Scholarship**); Tanglixin Scholarship; Gratitude for Modern Science Scholars Scholarship (10 selected university-wide, ranked first); Sichuan Outstanding Graduates; UESTC Outstanding Undergraduate Thesis Award; 10+ additional scholarships
- **Weighted GPA: 95.3/100 (broke school record)**; **Ranked 1st out of 643 students**; GPA 3.99/4.0
- Class Monitor for four years (led class to receive Outstanding Class Collective award for three consecutive years; personally awarded Top 10 Class Monitor and Outstanding Youth Leader); Vice Team Leader for Rural Social Practice (awarded Chengdu Excellent Rural Practice Team, University Special First Prize; selected as Excellent Practice Individual)

Work Experience

ByteDance - SEED - Heterogeneous Computing, Topseed, ByteDance Top Talent Program 2025.4 - Present

- **ByteMlperf**:
  - **Responsibility**: Framework construction and influence building for ByteMlperf
  - **Project Highlights**: ByteMlperf is ByteDance's platform to break information silos and promote deep understanding and adoption of domestic chips. The project consists of three components: 1) Analyze and summarize ByteDance's internal workload characteristics, open-source key workloads and scenarios, and publish existing challenges; 2) A complete benchmarking framework already deployed internally at ByteDance, standardizing test processes and enhancing data interpretability; 3) Deep profiling, analysis and comparison of domestic chips to help academia and industry better understand practical challenges and promote end-to-end development of domestic chips. (Primarily responsible for components 1 and 3)
  - **Expected Outcomes**: Corresponding framework and analysis results open-sourced and accepted at HPCA 2025 industry track as first author (**only 3-4 papers accepted annually**); Internal findings will inform chip selection and automatic framework development
- **Online Inference Optimization**:
  - **Responsibility**: Online service LLM inference optimization
  - **Project Highlights**: Responsible for online inference optimization, primarily focusing on Load Balance and Batching strategy optimization as foundation for overall self-developed inference framework optimization
  - **Current Outcomes**: 11.3% QPS throughput improvement achieved in production, significantly increased FTL achievement rate. Targeting 30% improvement. Results under preparation for SOSP 2026 submission

Arctic Xiongxin Information Technology Co., Ltd., Compiler Team Technical Lead + Chip Architect 2020.6 - Present

- **Chip Architect**:
  - **Responsibility**: Throughout four generations of Qiming series chips, I primarily led the design of multi-core interconnect architecture, single compute core microarchitecture, and instruction set architecture. Chips involved include: Qiming 930 (RDL packaging with 1 HUB Die and 4 NPU Dies, TSMC 12nm, 40TOPS total, released); Qiming 935 (1 HUB Die and 2 Side Dies (NPU/Multimedia), TSMC 12nm, testing completed, mass production planned); Qiming 940 (AI compute expansion and architecture refinement based on Qiming 935, currently in architecture design phase); Qiming 940-3D (3D DRAM-based LLM decode inference accelerator, architecture design phase)
  - **Challenges Addressed**: Having developed cutting-edge scheduling optimization strategies (ISCA 2023, HPCA 2024, DAC 2025, HPCA 2025 series), corresponding chip architecture and instruction-level architecture support are needed. I primarily define hardware support requirements from scheduling strategies and translate them into architecture specifications and microarchitecture designs
  - **Achievements**: Through design, the chip efficiently supports chiplet scalability, maximizes cost advantages of modular architecture, achieving high compute utilization in mainstream autonomous driving neural networks
- **Compiler Team Technical Lead**:
  - **Responsibility**: As compiler team technical lead, I led the team to successfully apply top-conference academic results to production, **spearheading** the construction of two generations of scalable compiler software stacks for 12nm large-scale DNN accelerators. During development, I primarily designed optimization module strategies by combining hardware characteristics and assigned these to team engineers for implementation. Responsible for technical problem solving and code review, and led hardware-software coupling to sync software-identified issues and required features to chip microarchitecture optimization
  - **Software Stack Features**: The second-generation stack underwent bottom-up reconstruction based on first-generation lessons, covering complete workflow from frontend IR parsing, operator fusion and graph optimization, data prefetching optimization, on-chip memory address optimization, IR generation optimization, instruction generation and optimization, to instruction-level simulator. The stack adopts modular design for easy optimization module upgrades and supports different compute capacity versions through chiplet combinations
  - **Achievements**: The compiler software stack successfully supports over 90% of ONNX operators and achieves high compute utilization in mainstream autonomous driving neural networks
  - **Current Work**: Currently, the team is fully supporting compilation and optimization work for Large Language Models (LLM), further improving performance and efficiency

Publications (*indicating co-first authors)

- **Jingwei Cai***, Dehao Kong, Hantao Huang, Zishan Jiang, Zixuan Ma, Qingyu Guo, Zhenxing Zhang, Guiming Shi, Mingyu Gao, Kaisheng Ma, Minghui Yu, "Characterizing Cloud-Native LLM Inference at ByteDance and Exposing Optimization Challenges and Opportunities for Future AI Accelerators" ***HPCA 2026 Industry Track (CCF-A, Top Computer Architecture Conference, Industry Track accepts only 3-4 papers from industry annually, with 20 submissions)***
  - **Research Innovation**: This work makes the following contributions: 1) Summarize actual workload characteristics and chip/system design challenges and opportunities using Doubao APP scenario as example. 2) Propose our evaluation system with three levels (model, operator, instruction) compared to MLPerf, enabling better interpretability of chip performance and discovery of application potential in different domains; this framework will be open-sourced with the paper. 3) Present architecture design challenges and limitations discovered during actual chip deployment, hoping these insights help academia identify valuable research directions and practically address current challenges in LLM inference chips. Additionally, hope this guides chip vendors to avoid these issues and design chips better suited for cloud inference scenarios
  - **Highlights**: This work consolidates extensive ByteDance accumulation on workload and chip requirements. Related framework will be open-sourced globally, working to establish more interpretable benchmark platforms
- **Jingwei Cai**, Zuotong Wu, Sen Peng, Yuchen Wei, Zhanhong Tan, Guiming Shi, Mingyu Gao, Kaisheng Ma, "Mapping and Architecture Co-exploration for Large-scale DNN Chiplet Accelerators" ***HPCA 2024 (CCF-A, Top Computer Architecture Conference); IEEE Available, Functional, Reproducible Badges; HPCA Travel Grant awarded***
  - **Research Innovation**: This work proposes the first mapping and architecture co-exploration framework for large-scale DNN chiplet accelerators and is the first to introduce cost factors into DNN Chiplet architecture exploration, filling a research gap in this field. This work also first clearly defines and deeply explores optimization space of layer pipeline spatial mapping, providing clear direction and foundation for future research. Through detailed framework exploration, this work reveals many counterintuitive insights on Chiplet-granularity architecture and spatial mapping scheduling. These insights not only broaden understanding of DNN Chiplet accelerator design and optimization but also establish solid foundation for future Chiplet accelerator research and applications
  - **Highlights**: This research received CCF-A HPCA 2024's "Distinguished Artifact Award (1/410), the first such achievement for Chinese teams at HPCA"
- **Jingwei Cai***, Yuchen Wei*, Zuotong Wu, Sen Peng, Kaisheng Ma, "Inter-layer Scheduling Space Definition and Exploration for Tiled Accelerators" ***ISCA 2023 (CCF-A, Top Computer Architecture Conference); ACM Available, Functional, Reproducible Badges; Also accepted by Chinasys 2023 with high score (445); ISCA Travel Grant awarded***

- **Research Content**: The first paper to define, explore and understand the inter-layer scheduling optimization space for Tiled/multi-core DNN accelerators, breaking the monopoly and severe limitations of traditional layer-pipeline and layer-sequential parallel modes
- **Product Deployment**: This work achieved **complete production deployment** in internal startup company (Arctic Xiongxin), on which was built end-to-end compiler framework covering frontend IR parsing, scheduling optimization, IR generation optimization, instruction generation and optimization throughout the complete workflow
- **Academic Impact**: This work received widespread attention from academia (MIT, Tsinghua, Peking University, Georgia Tech, Seoul National University, HKUST, Institute of Automation CAS, Institute of Computing CAS, Huazhong University of Science and Technology, etc.) and industry (Intel, etc.), achieving adoption at varying levels. We provided extensive internal support to further enhance tool impact
- **Highlights**: Tushar Krishna, computer architecture expert from Georgia Tech (author of Gem5 and Garnet, 16000+ Google Scholar citations, ISCA 2023 PC Chair), incorporated our framework into his course curriculum for student learning and use

- **Jingwei Cai**, Xuan Wang, Mingyu Gao, Sen Peng, Zijian Zhu, Yuchen Wei, Zuotong Wu, Kaisheng Ma, "Identifying, Exploring, and Understanding the DRAM Communication Scheduling Space for DNN Accelerators" ***HPCA 2025 (CCF-A, Top Computer Architecture Conference); IEEE Available, Functional, Reproducible Badges***
  - **Research Innovation**: As the gap between DRAM bandwidth and compute density grows increasingly large, optimizing DRAM communication has become a core task in compiler scheduling optimization. This work optimizes DRAM communication including fine-grained layer fusion optimization and data prefetching/delayed transmission optimization. Fine-grained layer fusion compared to traditional operator fusion saves on-chip buffer usage; data prefetching and delayed transmission are optimization dimensions previously largely overlooked, which this paper comprehensively analyzes and optimizes. Finally, the paper unifies both optimization dimensions, develops a framework for joint exploration, achieving 2.44x performance improvement and 48.2% energy reduction compared to SOTA ASPLOS COCCO
  - **Highlights**: Corresponding framework has completed compiler development for a large-scale autonomous driving chip with complete production deployment. Chip has returned from foundry, mass production expected next year
- Yuchen Wei*, **Jingwei Cai***, Mingyu Gao, Sen Peng, Zuotong Wu, Guiming Shi, Kaisheng Ma, "Discovering and Exploiting Untapped Buffer Resources in Many-Core DNN Accelerators" ***DAC 2025 (CCF-A EDA Top Conference)***. This paper identifies inherent Buffer Underutilized phenomenon in layer pipeline mapping, proposes novel buffer scheduling strategies to enable more on-chip reuse opportunities, achieving significant performance and energy efficiency improvements
- Guiming Shi, Zhanhong Tan, Dapeng Cao, **Jingwei Cai**, Wuke Zhang, Yifu Wu, Kaisheng Ma, "A 28nm 68MOPS 0.18J/Op Paillier Homomorphic Encryption Processor with Bit-Serial Sparse Ciphertext Computing" ***ISSCC 2023*** (Top Chip Design Conference)
- Guiming Shi, Yi Li, Xueqiang Wang, Zhanhong Tan, Dapeng Cao, **Jingwei Cai**, Yuchen Wei, Zehua Li, Yifu Wu, Wuke Zhang, Wei Xu, and Kaisheng Ma, "PHEP: Paillier Homomorphic Encryption Processors for Privacy-Preserving Applications in Cloud Computing" ***HOT CHIPS 2023*** (Top Chip Design Conference)

Patents

- **A Neural Network Mapping Framework for Multi-chiplet Systems**, First Inventor, Invention Patent, Filing Date December 2023 (Pending)
- **On-chip Cache Allocation Method and Related Device for Deep Neural Network Multi-core Accelerator**, First Inventor, Invention Patent, Filing Date December 2023 (Pending)
- **A Tiled Accelerator Resource Allocation Method and System**, First Inventor, Invention Patent, Filing Date December 2023 (Pending)

Ongoing Projects (All as Technical Lead or Spearheader)

- **Heterogeneous Computing Unit Parallelization Optimization for DNN Accelerators, Oct 2024-Present**: Address inherent mutual waste phenomenon between GEMM resources and Vector resources in DNN accelerators through architecture and fine-grained operator parallelization optimization to reduce resource idle time and improve performance. Currently underway, targeting MICRO 2025 submission (CCF-A Computer Architecture Top Conference); **Will achieve production deployment at Arctic Xiongxin**
- **3D DRAM-based LLM Inference Accelerator Architecture and Compiler Research, May 2024-Present**: LLM inference pushes bandwidth requirements to unprecedented levels. With recent advances in 3D technology, 3D DRAM is maturing, but compared to traditional DRAM, its inherent distributed characteristics pose new requirements for architecture and scheduling. How to design appropriate granularity and scheduling approaches to fully leverage 3D DRAM bandwidth potential is a critical question and project focus. Simultaneously designing actual 3D DRAM-based LLM decode accelerator
- **Silicon Photonic Interconnected Wafer-scale Computing System Design and Research, Oct 2022-Present**: Since 2022, I have served as group representative, deeply participating in the "14th Five-Year Plan National Key R&D Program Key Project" - Wafer-scale Silicon Photonic Interconnected Computing System project. This project integrates 48 compute chiplets and 4 silicon photonic switch chiplets using advanced packaging technology, aiming to achieve peta-scale computing capacity, with enormous technical challenges
  - **Responsibility**: Led research teams from Tsinghua and Xi'an Research Institute composed of graduate students and engineers, fully responsible for computing system architecture and compiler development, and played key roles in overall packaging architecture design
  - **Innovative Technical Solutions**: Proposed a series of innovative technical solutions driving smooth project progress and effectively addressing core challenges. Particularly, based on silicon photonic interconnect characteristics, proposed heterogeneous

chiplet computing architecture and multi-level hybrid parallel scheduling strategies, fully leveraging advantages of compute and photonic switch chiplets, significantly improving overall system performance

- **Project Progress**: Currently in final tape-out phase, related results expected to submit to *Nature Photonics* next year